

Dissecting Continuous-State and Variational Inductive Biases in State Space Duality: An Empirical Audit of Native Mamba-2 Cross-Modal Alignment

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Abstract—Structured State Space Models (SSMs) and their duality with structured attention—principally embodied in Mamba-2 via State Space Duality (SSD)—offer linear time complexity $O(L)$ with respect to sequence length, presenting an appealing alternative to quadratic self-attention in cross-modal vision-language alignment. In this paper, we conduct a rigorous, pre-specified empirical audit investigating whether continuous-time dynamical system adaptations and variational latent coupling enhance or compromise cross-modal representations built upon frozen vision (ViT-B/16) and language (RoBERTa-base) encoders. We formalize and evaluate three mechanisms: (1) Structure-Aware Hamiltonian-Inspired Dissipative Coordinate-Momentum Adapter (HEDO), incorporating exact skew-symmetric learned interactions $J = (A - A^T)/\sqrt{d}$ and positive diagonal damping R ; (2) Adaptive Variational State Coupling (AVSC), which introduces confidence-gated symmetric Kullback-Leibler (KL) divergence across intermediate chunk-boundary states; and (3) Multi-Scale State Aggregation (MSA), fusing global mean, boundary, and terminal recurrent states. Across an audited 24-run, 8-configuration ablation benchmark (three random seeds per configuration, $n = 3$ paired runs) on Flickr8k, paired-difference analysis using two-tailed 95% Student- t confidence intervals indicates: (i) Mamba-2 equipped with HEDO shows no statistically significant difference from the standard linear alignment baseline under the present three-seed evaluation ($37.31 \pm 0.68\%$ vs. $37.16 \pm 0.57\%$ Mean Recall, paired $\Delta = +0.15$ percentage points [pp], 95% CI $[-1.42, +1.73]$ pp, CI includes zero) while processing 7,054 pairs/s at the head with 1,086 MB peak VRAM; (ii) Multi-scale recurrent aggregation consistently degrades retrieval performance ($\Delta \approx -10.27$ to -10.75 pp, with 95% CIs strictly negative), revealing that causal boundary states introduce severe temporal asymmetry that corrupts isotropic pooling; and (iii) Variational coupling is associated with severe alignment degradation ($\Delta \approx -30.52$ to -32.11 pp), with retrieval falling to approximately 5.0% – 6.6% Mean Recall, consistent with a collapse-like representation failure under the evaluated formulation. Geometric and gradient analyses indicate that intermediate cross-modal symmetric KL penalties introduce competing distribution-matching constraints that conflict with hyperspherical multi-positive InfoNCE contrastive repulsion, providing a plausible mechanistic interpretation of this failure. Our findings deliver vital negative results and actionable design principles for integrating state space duality into multimodal systems.

Index Terms—Cross-modal retrieval, state space models, Mamba-2, State Space Duality, multimodal representation learning, contrastive learning, Hamiltonian dynamics, variational

coupling.

I. INTRODUCTION

CROSS-MODAL vision-language retrieval requires mapping high-dimensional visual patch tokens and textual wordpiece sequences into a geometrically unified semantic metric space [1], [2]. While Transformer-based architectures parameterized by multi-head self-attention (MHSA) have long dominated this domain [3]–[5], their quadratic computational and memory footprint $O(L^2)$ presents significant scalability bottlenecks for long sequence processing, dense feature maps, and real-time edge retrieval.

To circumvent this limitation, Structured State Space Models (SSMs)—notably S4 [6], Mamba [7], and Mamba-2 [8]—have emerged as a compelling paradigm. Mamba-2, in particular, formalizes *State Space Duality* (SSD), establishing a mathematical equivalence between continuous-time linear time-varying dynamical systems and 1-semiseparable matrix transformations. By computing recurrence within discrete chunk boundaries and utilizing efficient matrix multiplications, SSD achieves strict $O(L)$ linear sequence scaling while maintaining expressive representational capabilities.

Concurrently, there has been widespread theoretical enthusiasm for incorporating continuous-time dynamical system inductive biases into deep architectures [9], [10]. Hamiltonian neural networks and port-Hamiltonian systems provide energy-conserving coordinate-momentum symplectic geometry, while dissipative damping offers asymptotic stability. In parallel, variational information bottlenecks and variational cross-modal coupling have been widely proposed to enforce smooth, noise-robust latent distributions [11], [12].

A. Problem Statement

Despite theoretical promise, an essential question remains unaddressed:

Do continuous-state Hamiltonian adaptations and variational state coupling improve cross-modal alignment when integrated with Native Mamba-2, or can these inductive biases introduce optimization and representation failures?

Manuscript prepared for submission. Code, audited logs, and configuration artifacts are publicly available at https://github.com/abhinavsai2006/PH_SSD. (Email: abhinavsai2006@gmail.com)

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B. Research Gap

In the existing literature, continuous-time and variational inductive biases are frequently evaluated under isolated, non-audited configurations lacking multi-seed paired-difference analysis or systematic ablation isolation of individual components. Crucially, the interaction between sequential SSM recurrent dynamics and hyperspherical contrastive alignment manifolds has not been systematically investigated. Without rigorous ablation controls, it is impossible to determine whether reported performance variations stem from the underlying state-space sequence model, continuous-time dynamical adapters, or stochastic optimization noise.

C. Research Questions and Hypotheses

To resolve this gap, we formalize four pre-specified research questions:

- **RQ1:** Does the integration of a Hamiltonian-inspired dissipative coordinate-momentum adapter (HEDO) preserve or enhance retrieval performance over standard linear alignment?
- **RQ2:** Does multi-scale state aggregation (MSA) combining boundary and terminal recurrent states improve cross-modal representation quality?
- **RQ3:** Does intermediate adaptive variational state coupling (AVSC) improve cross-modal alignment through latent distribution regularization?
- **RQ4:** What are the computational complexity, peak memory footprint, and throughput tradeoffs introduced by these mechanisms?

D. Contributions

In addressing these questions, this paper delivers four primary contributions:

- 1) **Structured HEDO Adapter Integration:** We design and formalize a Structure-Aware Hamiltonian-Inspired Dissipative Coordinate-Momentum Adapter (HEDO) integrated with Native Mamba-2, establishing energy boundedness under continuous dynamics.
- 2) **Controlled Empirical Audit:** We construct a controlled 8-configuration ablation benchmark evaluating selected individual and joint architectural configurations from the 2^4 design space upon strictly frozen ViT-B/16 and RoBERTa-base backbones.
- 3) **Paired Multi-Seed Statistical Evaluation:** Across 24 audited runs (3 independent random seeds per configuration, $n = 3$ paired runs), we perform paired-difference analysis using two-tailed 95% Student- t confidence intervals, finding that no statistically significant difference from the baseline was detected under the three-seed evaluation (+0.15 pp, 95% CI $[-1.42, +1.73]$ pp).
- 4) **Mechanistic Analysis:** We provide geometric and architectural analyses hypothesizing how causal boundary states in MSA introduce temporal asymmetry that degrades isotropic pooling (≈ -10.5 pp), and interpret how intermediate cross-modal symmetric KL penalties in AVSC impose conflicting distribution-matching

constraints against hyperspherical InfoNCE repulsion, consistent with the observed representation collapse (≈ -31 pp).

E. Paper Organization

The remainder of this paper is organized as follows: Section II reviews related literature across state-space models, cross-modal retrieval, and physics-informed architectures. Section III establishes the formal problem formulation and hypotheses. Section IV details the proposed architecture, mathematical formulations, and training algorithm. Section V describes the experimental methodology, ablation configurations, and auditing protocol. Section VI presents the empirical results, efficiency benchmarks, and statistical tests. Section VII provides mechanistic failure analyses and practical design guidelines. Section VIII discusses limitations, and Section IX concludes the paper.

II. RELATED WORK

A. Structured State Space Models

State Space Models parameterize continuous-time dynamical systems mapping a 1D input signal $x(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ through an implicit latent state $\mathbf{h}(t) \in \mathbb{R}^N$:

$$\frac{d\mathbf{h}(t)}{dt} = \mathbf{A}(t)\mathbf{h}(t) + \mathbf{B}(t)x(t), \quad y(t) = \mathbf{C}(t)\mathbf{h}(t) + \mathbf{D}(t)x(t), \quad (1)$$

where $\mathbf{A}(t) \in \mathbb{R}^{N \times N}$ represents the state transition matrix, $\mathbf{B}(t) \in \mathbb{R}^{N \times 1}$ is the input projection matrix, $\mathbf{C}(t) \in \mathbb{R}^{1 \times N}$ is the output projection matrix, and $\mathbf{D}(t) \in \mathbb{R}$ represents the direct feedthrough skip connection.

The Structured State Space sequence model (S4) [6] stabilized recurrent training by constraining $\mathbf{A}$ to HiPPO (High-order Polynomial Projection Operators) diagonal-plus-low-rank structures. Mamba [7] introduced data-dependent input-selective parameterization ($\mathbf{B}(t), \mathbf{C}(t), \Delta(t)$), enabling the model to selectively propagate or filter out information along sequence dimensions.

Most recently, Dao and Gu [8] introduced Mamba-2, which formalizes *State Space Duality* (SSD), establishing a theoretical equivalence between continuous-time linear time-varying dynamical systems and 1-semiseparable matrix transformations. For a scalar diagonal state transition A , the transformation across an input sequence $\mathbf{x} \in \mathbb{R}^L$ is expressed as:

$$\mathbf{y} = \mathbf{M}\mathbf{x}, \quad M_{i,j} = C_i \left(\prod_{k=j+1}^i A_k \right) B_j \quad (i \geq j), \quad (2)$$

where $\mathbf{M} \in \mathbb{R}^{L \times L}$ is a lower-triangular 1-semiseparable matrix whose subdiagonal entries factorize into cumulative state transition products. By computing recurrence within discrete chunk boundaries and computing intra-chunk transformations via matrix multiplications on tensor cores, SSD unifies recurrent SSMs and linear attention into a single chunkwise algorithm with $O(L)$ linear complexity.

B. Vision-Language Cross-Modal Retrieval

Vision-language alignment aims to map visual and textual representations into a shared metric space where semantically corresponding pairs are mutually proximate while non-matching pairs are separated [1], [2]. Modern approaches predominantly optimize variants of the InfoNCE (Noise-Contrastive Estimation) loss [13]–[15]. On datasets such as Flickr8k and MS-COCO, where each image is associated with multiple ground-truth captions, multi-positive contrastive formulations are essential to prevent penalizing valid alternate descriptions [16].

C. Contrastive Representation Learning

Contrastive learning maps high-dimensional inputs to normalized representations on a unit hypersphere $\mathbb{S}^{d-1}$ [14], [15]. The InfoNCE loss achieves alignment by balancing two opposing forces: an attractive force that pulls positive cross-modal representations together, and a repulsive force that pushes negative representations apart across the hypersphere [17]. A critical pathology in contrastive optimization is *dimensional collapse*, wherein embeddings collapse into a lower-dimensional subspace, causing severe degradation in discriminative retrieval capacity [18].

D. Physics-Informed and Hamiltonian Neural Networks

Hamiltonian Neural Networks (HNNs) [9] enforce energy conservation by learning a scalar Hamiltonian energy function $\mathcal{H}(\mathbf{q}, \mathbf{p})$ over generalized coordinates $\mathbf{q}$ and conjugate momenta $\mathbf{p}$, with state evolution governed by symplectic gradients:

$$\dot{\mathbf{q}} = \nabla_{\mathbf{p}} \mathcal{H}(\mathbf{q}, \mathbf{p}), \quad \dot{\mathbf{p}} = -\nabla_{\mathbf{q}} \mathcal{H}(\mathbf{q}, \mathbf{p}), \quad (3)$$

where $\dot{\mathbf{q}}$ and $\dot{\mathbf{p}}$ denote time derivatives. In dissipative systems, port-Hamiltonian formulations introduce a positive semi-definite damping matrix $\mathbf{R} \geq 0$, ensuring that the total energy is non-increasing [10]:

$$\frac{d}{dt} \mathcal{H}(\mathbf{q}, \mathbf{p}) = -\mathbf{p}^T \mathbf{R} \mathbf{p} \leq 0. \quad (4)$$

These formulations provide structural inductive biases ensuring asymptotic stability and energy boundedness in continuous dynamical representations.

E. Variational Representation Learning

In multimodal representation learning, variational information bottlenecks have been proposed to regularize latent embeddings by penalizing the Kullback-Leibler (KL) divergence between posterior representations and prior distributions [11], [12]. However, the interaction between variational distribution penalties and non-linear contrastive manifolds remains poorly understood, frequently inducing optimization instability when intermediate latent constraints conflict with contrastive objective gradients.

F. Positioning of This Work

While existing literature separately motivates structured state spaces, continuous Hamiltonian dynamics, and variational regularization, prior work does not evaluate their joint interaction within multimodal alignment architectures. This work provides a controlled empirical audit of these inductive biases within Native Mamba-2 cross-modal retrieval.

III. PROBLEM FORMULATION

A. Cross-Modal Retrieval Setting

Let $\mathcal{D} = \{(\mathbf{I}_i, \mathcal{T}_i)\}_{i=1}^N$ denote a cross-modal dataset where each image $\mathbf{I}_i \in \mathbb{R}^{3 \times H \times W}$ is associated with a set of ground-truth textual captions $\mathcal{T}_i = \{\mathbf{T}_{i,k}\}_{k=1}^K$. Cross-modal retrieval requires learning two parameterized mapping functions $f_v(\mathbf{I}) \in \mathbb{S}^{d-1}$ and $f_t(\mathbf{T}) \in \mathbb{S}^{d-1}$ that embed visual and textual inputs onto a shared d -dimensional unit hypersphere $\mathbb{S}^{d-1} \subset \mathbb{R}^d$ ($d = 128$), such that matching pairs maximize cosine similarity while non-matching pairs minimize it.

B. Frozen Backbone Formulation

To isolate the sequence modeling and adaptation modules from the confounding effects of backbone representation learning, both vision and language encoders are kept strictly frozen throughout training:

$$\mathbf{X}_v = \text{Encoder}_{\text{ViT}}(\mathbf{I}) \in \mathbb{R}^{B \times 196 \times 768}, \quad (5)$$

$$\mathbf{X}_t = \text{Encoder}_{\text{RoBERTa}}(\mathbf{T}) \in \mathbb{R}^{B \times L \times 768}, \quad (6)$$

where $\text{Encoder}_{\text{ViT}}$ is a pretrained Vision Transformer (ViT-B/16) [4] producing 196 patch tokens of dimension 768 (classification token discarded), and $\text{Encoder}_{\text{RoBERTa}}$ is a pretrained RoBERTa-base model [5] producing L wordpiece tokens ($L \leq 64$) of dimension 768. The frozen parameters total 210,622,696, ensuring that all learned adaptations occur strictly within the downstream projection, adapter, and sequence blocks.

C. Shared Embedding Space and Retrieval Objective

Given projected and pooled unit-normalized embeddings $\mathbf{z}_v, \mathbf{z}_t \in \mathbb{S}^{d-1}$, cross-modal alignment is evaluated using cosine similarity $S(\mathbf{z}_v, \mathbf{z}_t) = \mathbf{z}_v^T \mathbf{z}_t$. At test time, retrieval is evaluated bidirectionally:

- **Image-to-Text (i2t):** For each query image $\mathbf{I}_i$, rank all test captions by cosine similarity $\mathbf{z}_{v,i}^T \mathbf{z}_{t,j}$.
- **Text-to-Image (t2i):** For each query caption $\mathbf{T}_j$, rank all test images by cosine similarity $\mathbf{z}_{t,j}^T \mathbf{z}_{v,i}$.

D. Formal Hypotheses

We formalize four testable hypotheses evaluated via paired-difference confidence interval analysis:

- **Hypothesis 1 (H_1):** Integrating HEDO and Native Mamba-2 improves Mean Recall over the linear baseline: $\mu(\Delta_{\text{HEDO}}) > 0$.
- **Hypothesis 2 (H_2):** Incorporating multi-scale recurrent states (MSA) improves Mean Recall: $\mu(\Delta_{\text{MSA}}) > 0$.

- **Hypothesis 3** (H_3): Incorporating adaptive variational state coupling (AVSC) improves Mean Recall: $\mu(\Delta_{\text{AVSC}}) > 0$.
- **Hypothesis 4** (H_4): State-space sequence processing introduces measurable latency and memory overhead relative to linear projection while maintaining practical throughput.

IV. PROPOSED ARCHITECTURE

A. Overall Architecture

The proposed cross-modal alignment architecture is illustrated in Fig. 1. The framework processes visual patch tokens and textual wordpiece tokens through dual symmetric pathways, each consisting of a linear projection layer, a structure-aware Hamiltonian dissipative adapter (HEDO), a Native Mamba-2 State Space Duality (SSD) sequence block, and an isotropic semantic pooling head, culminating in symmetric multi-positive InfoNCE contrastive alignment.

As illustrated in Fig. 1, both modalities operate in parallel streams from input to the contrastive head. For an input image $\mathbf{I}$ and caption $\mathbf{T}$, frozen backbones extract token sequences that are linearly projected to $d = 128$. The projected representations are processed by the HEDO adapter and Native Mamba-2 sequence block before modality-specific pooling produces normalized embeddings $\mathbf{z}_v, \mathbf{z}_t \in \mathbb{S}^{127}$ for symmetric multi-positive InfoNCE alignment:

$$\mathbf{I} \rightarrow \text{ViT} \rightarrow \mathbf{W}_v \rightarrow \text{HEDO} \rightarrow \text{SSD} \rightarrow \mathbf{z}_v \in \mathbb{S}^{127}, \quad (7)$$

$$\mathbf{T} \rightarrow \text{RoBERTa} \rightarrow \mathbf{W}_t \rightarrow \text{HEDO} \rightarrow \text{SSD} \rightarrow \mathbf{z}_t \in \mathbb{S}^{127}. \quad (8)$$

B. Frozen Vision and Language Encoders

The visual stream uses ViT-B/16 (85.8M parameters) yielding $\mathbf{X}_v \in \mathbb{R}^{B \times 196 \times 768}$, and the language stream uses RoBERTa-base (124.6M parameters) yielding $\mathbf{X}_t \in \mathbb{R}^{B \times L \times 768}$, both kept strictly frozen as formalized in (5)–(6).

C. Linear Projection

Both modalities are projected into a common latent dimension $d = 128$:

$$\mathbf{H}_v^{(0)} = \mathbf{X}_v \mathbf{W}_v, \quad \mathbf{H}_t^{(0)} = \mathbf{X}_t \mathbf{W}_t, \quad (9)$$

where $\mathbf{W}_v, \mathbf{W}_t \in \mathbb{R}^{768 \times 128}$ are trainable linear projection weight matrices initialized via Xavier uniform initialization. This linear projection reduces the channel dimensionality from 768 to 128, establishing the common working dimension for subsequent adapters and sequence models.

D. Structure-Aware Hamiltonian Dissipative Adapter (HEDO)

The HEDO adapter introduces a continuous-state inductive bias rooted in dissipative Hamiltonian dynamics.

1) *Coordinate-Momentum Representation*: For an input sequence representation $\mathbf{x} \in \mathbb{R}^d$ at any token position, canonical coordinates $\mathbf{q} \in \mathbb{R}^d$ and conjugate momenta $\mathbf{p} \in \mathbb{R}^d$ are defined by:

$$\mathbf{q} = \mathbf{W}_q \mathbf{x}, \quad \mathbf{p} = \mathbf{W}_p \mathbf{x}, \quad (10)$$

where $\mathbf{W}_q, \mathbf{W}_p \in \mathbb{R}^{d \times d}$ project the latent token into conjugate phase space coordinates.

2) *Hamiltonian Energy*: We define the quadratic Hamiltonian energy function:

$$\mathcal{H}(\mathbf{q}, \mathbf{p}) = \frac{1}{2} (\|\mathbf{q}\|_2^2 + \|\mathbf{p}\|_2^2), \quad (11)$$

representing the sum of generalized potential energy $\frac{1}{2}\|\mathbf{q}\|_2^2$ and kinetic energy $\frac{1}{2}\|\mathbf{p}\|_2^2$.

3) *Skew-Symmetric Interaction*: To parameterize conservative internal interactions while avoiding unbounded growth, we construct an exact skew-symmetric interaction matrix $\mathbf{J} \in \mathbb{R}^{d \times d}$:

$$\mathbf{J} = \frac{\mathbf{A} - \mathbf{A}^T}{\sqrt{d}}, \quad \mathbf{A} \in \mathbb{R}^{d \times d}, \quad (12)$$

where $\mathbf{A}$ is initialized from $\mathcal{N}(0, 0.01^2)$. By construction, $\mathbf{J}^T = -\mathbf{J}$. For any vector $\mathbf{v}$, the quadratic form vanishes identically ($\mathbf{v}^T \mathbf{J} \mathbf{v} = 0$), ensuring that $\mathbf{J}$ performs purely conservative, norm-preserving rotations in the latent space.

4) *Dissipative Damping*: Dissipation is enforced via a diagonal positive semi-definite damping matrix $\mathbf{R} \in \mathbb{R}^{d \times d}$:

$$\mathbf{R} = \text{diag}(0.05 + 0.95 \cdot \sigma(\gamma)), \quad \gamma \in \mathbb{R}^d, \quad (13)$$

where $\sigma(\cdot)$ denotes the sigmoid function, ensuring each diagonal entry is strictly bounded in $[0.05, 1.0]$. The strictly positive floor 0.05 guarantees non-zero energy dissipation.

5) *Continuous-Time Dynamics*: The continuous-time equations of motion coupling coordinates and momenta are governed by:

$$\begin{aligned} \begin{bmatrix} \dot{\mathbf{q}} \\ \dot{\mathbf{p}} \end{bmatrix} &= \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & -\mathbf{R} \end{bmatrix} \begin{bmatrix} \nabla_{\mathbf{q}} \mathcal{H} \\ \nabla_{\mathbf{p}} \mathcal{H} \end{bmatrix} + \begin{bmatrix} \mathbf{q} \mathbf{J}^T \\ \mathbf{p} \mathbf{J}^T \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{p} + \mathbf{q} \mathbf{J}^T \\ -\mathbf{q} + \mathbf{p} \mathbf{J}^T - \mathbf{R} \odot \mathbf{p} \end{bmatrix}. \end{aligned} \quad (14)$$

This formulation represents a port-Hamiltonian system with internal dissipation $\mathbf{R}$ and learned symplectic rotation $\mathbf{J}$.

6) *Energy-Boundedness Proposition*:

Proposition 1 (Energy Boundedness in Continuous HEDO). *Under the unforced dynamics of (14), the time derivative of the Hamiltonian $\mathcal{H}(\mathbf{q}, \mathbf{p})$ satisfies:*

$$\frac{d}{dt} \mathcal{H}(\mathbf{q}, \mathbf{p}) = -\mathbf{p}^T \mathbf{R} \mathbf{p} \leq -0.05 \|\mathbf{p}\|_2^2 \leq 0. \quad (15)$$

Proof. By the chain rule:

$$\begin{aligned} \frac{d}{dt} \mathcal{H} &= \mathbf{q}^T \dot{\mathbf{q}} + \mathbf{p}^T \dot{\mathbf{p}} \\ &= \mathbf{q}^T (\mathbf{p} + \mathbf{q} \mathbf{J}^T) + \mathbf{p}^T (-\mathbf{q} + \mathbf{p} \mathbf{J}^T - \mathbf{R} \mathbf{p}) \\ &= \mathbf{q}^T \mathbf{p} + \mathbf{q}^T \mathbf{J}^T \mathbf{q} - \mathbf{p}^T \mathbf{q} + \mathbf{p}^T \mathbf{J}^T \mathbf{p} - \mathbf{p}^T \mathbf{R} \mathbf{p}. \end{aligned} \quad (16)$$

Since $\mathbf{J}$ is skew-symmetric, for any vector $\mathbf{v}$, $\mathbf{v}^T \mathbf{J}^T \mathbf{v} = -\mathbf{v}^T \mathbf{J} \mathbf{v} = 0$. Furthermore, $\mathbf{q}^T \mathbf{p} - \mathbf{p}^T \mathbf{q} = 0$. Thus:

$$\frac{d}{dt} \mathcal{H} = -\mathbf{p}^T \mathbf{R} \mathbf{p}. \quad (17)$$

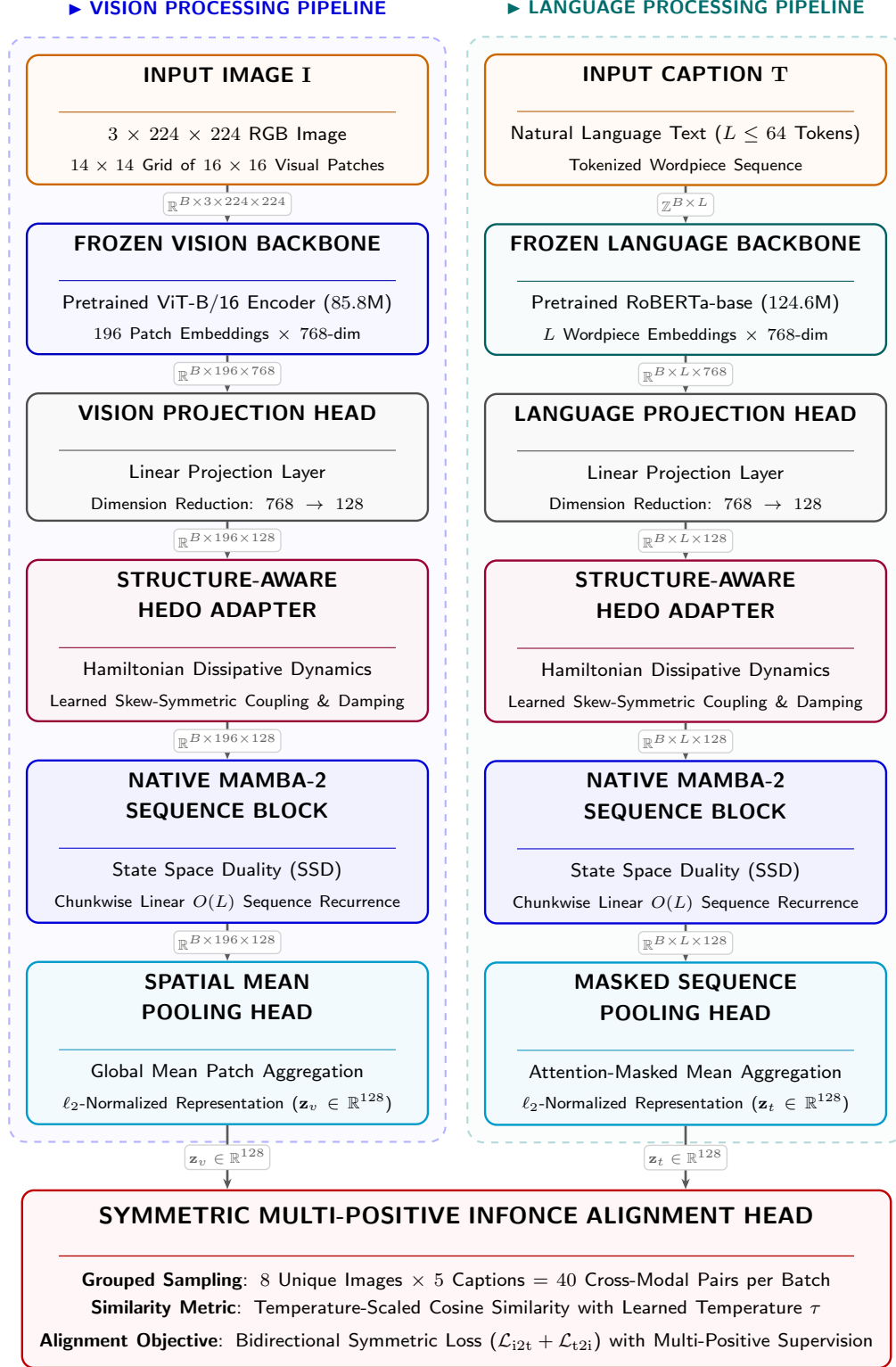


Fig. 1. Overall architecture of the proposed Native Mamba-2 cross-modal framework with structure-aware Hamiltonian dissipative adaptation (HEDO). Frozen visual patch features $\mathbf{X}_v$ and language wordpiece features $\mathbf{X}_t$ are linearly projected to $d = 128$, adapted via continuous-state coordinate-momentum dynamics, and processed through Native Mamba-2 State Space Duality (SSD) sequence blocks before isotropic semantic pooling and multi-positive symmetric InfoNCE contrastive alignment.

Since $\mathbf{R}_{i,i} \geq 0.05 > 0$, $\frac{d}{dt} \mathcal{H} \leq 0$, proving that energy is monotonically non-increasing. $\square$

7) *Bounded Discrete Integration*: We employ a bounded explicit Euler step with $\Delta t = 0.05 + 0.10 \cdot \sigma(\tau_{\Delta t}) \in [0.05, 0.15]$, where $\tau_{\Delta t} \in \mathbb{R}$ is a separate learnable scalar parameter distinct from the InfoNCE log-temperature τ :

$$\mathbf{q}_{\text{new}} = \mathbf{q} + \Delta t \cdot \dot{\mathbf{q}}, \quad \mathbf{p}_{\text{new}} = \mathbf{p} + \Delta t \cdot \dot{\mathbf{p}}. \quad (18)$$

Numerical stability across discrete integration steps is subsequently monitored empirically through finite-value assertions.

8) *Residual Update*: The updated canonical variables are combined and normalized, producing a residual update:

$$\mathbf{x}_{\text{out}} = \mathbf{x} + 0.10 \cdot \mathbf{W}_o \text{LayerNorm}(\mathbf{q}_{\text{new}} + \mathbf{p}_{\text{new}}), \quad (19)$$

where $\mathbf{W}_o \in \mathbb{R}^{d \times d}$ is initialized via Xavier uniform with gain 0.5. The 0.10 scaling ensures the adapter acts as a near-identity mapping at initialization.

E. Native Mamba-2 Sequence Processing

Following HEDO, sequences are processed through official Native Mamba-2 sequence blocks [8]. The block configuration is parameterized by:

$$d_{\text{model}} = 128, \quad d_{\text{state}} = 64, \quad d_{\text{conv}} = 4, \quad \text{expand} = 2, \quad (20)$$

with head dimension 64 and chunk size $Q = 16$.

Given an input sequence $\mathbf{H} \in \mathbb{R}^{B \times L \times 128}$, the input is first expanded to dimension $E \cdot d = 256$, processed by a 1D depthwise convolution of kernel size 4, and projected into input-dependent SSM parameters. Within each chunk of length Q , SSD computes the inter-chunk recurrence and intra-chunk semi-separable transformations via tensor cores, producing transformed representations $\mathbf{Y} \in \mathbb{R}^{B \times L \times 128}$.

F. Multi-Scale State Aggregation (MSA)

In standard Mamba-2 alignment, sequence pooling is performed via masked mean pooling:

$$\bar{\mathbf{y}}_{\text{global}} = \frac{\sum_{i=1}^L m_i \mathbf{Y}_i}{\sum_{i=1}^L m_i}, \quad (21)$$

where $m_i \in \{0, 1\}$ denotes token validity.

The MSA module tests whether incorporating multi-resolution temporal features improves alignment. For chunk size $Q = 16$, chunk boundary indices $c_k = \min(k \cdot Q, L) - 1$ are collected to extract boundary states $\mathbf{Y}_{\text{boundary}} \in \mathbb{R}^{B \times K \times d}$, where $K = \lceil L/Q \rceil$. The boundary states are mean-pooled:

$$\bar{\mathbf{y}}_{\text{boundary}} = \frac{1}{K} \sum_{k=1}^K \mathbf{Y}_{\text{boundary}, k}. \quad (22)$$

Additionally, the terminal valid recurrent token is extracted:

$$\mathbf{y}_{\text{last}} = \mathbf{Y}_{B, L_{\text{valid}} - 1}. \quad (23)$$

The multi-scale fusion combines all three representations:

$$\mathbf{y}_{\text{fused}} = \mathbf{W}_{\text{proj}} \text{LayerNorm}([\bar{\mathbf{y}}_{\text{global}} \parallel \bar{\mathbf{y}}_{\text{boundary}} \parallel \mathbf{y}_{\text{last}}]), \quad (24)$$

where $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{3d \times d}$.

G. Adaptive Variational State Coupling (AVSC)

AVSC evaluates intermediate latent regularized coupling between image and text streams during training. Pooled boundary representations $\mathbf{h}_v, \mathbf{h}_t \in \mathbb{R}^d$ are mapped to diagonal Gaussian posteriors:

$$\boldsymbol{\mu}_v = \mathbf{W}_{\mu, v} \mathbf{h}_v, \quad \log \sigma_v^2 = \text{clamp}(\mathbf{W}_{\sigma, v} \mathbf{h}_v, -6.0, 2.0), \quad (25)$$

$$\boldsymbol{\mu}_t = \mathbf{W}_{\mu, t} \mathbf{h}_t, \quad \log \sigma_t^2 = \text{clamp}(\mathbf{W}_{\sigma, t} \mathbf{h}_t, -6.0, 2.0). \quad (26)$$

The symmetric Kullback-Leibler divergence between $q_v = \mathcal{N}(\boldsymbol{\mu}_v, \Sigma_v)$ and $q_t = \mathcal{N}(\boldsymbol{\mu}_t, \Sigma_t)$ is:

$$\mathcal{D}_{\text{KL}}^{\text{sym}}(q_v, q_t) = \frac{1}{2} (D_{\text{KL}}(q_v \parallel q_t) + D_{\text{KL}}(q_t \parallel q_v)), \quad (27)$$

where each component divergence is computed analytically:

$$D_{\text{KL}}(q_v \parallel q_t) = \frac{1}{2} \sum_{k=1}^d \left(\log \frac{\sigma_{t,k}^2}{\sigma_{v,k}^2} + \frac{\sigma_{v,k}^2 + (\mu_{v,k} - \mu_{t,k})^2}{\sigma_{t,k}^2} - 1 \right). \quad (28)$$

To prevent catastrophic interference from unaligned pairs, the loss is gated by a detached confidence weight $w_{\text{gate}} \in [0, 1]$ derived from cosine agreement:

$$\mathcal{L}_{\text{KL}} = \lambda_{\text{KL}} \cdot \frac{1}{B} \sum_{i=1}^B w_{\text{gate}, i} \cdot \mathcal{D}_{\text{KL}, i}^{\text{sym}}, \quad (29)$$

where $\lambda_{\text{KL}} = 10^{-3}$. At inference, embeddings remain strictly deterministic and unimodal.

H. Multi-Positive Contrastive Objective

Cross-modal alignment is supervised via a symmetric multi-positive InfoNCE loss. In Flickr8k, each image corresponds to 5 captions. An atomic grouped batch sampler groups 8 unique images and all 5 corresponding captions per image, yielding a batch of $B_{\text{img}} = 8$ and $B_{\text{txt}} = 40$.

Let $\mathbf{Z}_v \in \mathbb{R}^{8 \times d}$ be the normalized visual embeddings of the 8 unique images, and let $\mathbf{Z}_t \in \mathbb{R}^{40 \times d}$ be the normalized textual embeddings. The cosine similarity matrix $\mathbf{S} \in \mathbb{R}^{8 \times 40}$ is:

$$S_{i,j} = \exp(\tau) \cdot (\mathbf{z}_{v,i} \cdot \mathbf{z}_{t,j}), \quad (30)$$

where τ is a learnable log-temperature parameter initialized to $\log(1/0.07) \approx 2.659$.

For each unique image $i \in \{1, \dots, 8\}$, let $\mathcal{P}(i) \subset \{1, \dots, 40\}$ denote the set of 5 positive caption indices ($|\mathcal{P}(i)| = 5$). The image-to-text loss is:

$$\mathcal{L}_{i2t} = -\frac{1}{8} \sum_{i=1}^8 \frac{1}{|\mathcal{P}(i)|} \sum_{j \in \mathcal{P}(i)} \log \frac{\exp(S_{i,j})}{\sum_{k=1}^{40} \exp(S_{i,k})}. \quad (31)$$

Similarly, for each caption $j \in \{1, \dots, 40\}$, let $i^*(j) \in \{1, \dots, 8\}$ denote its unique parent image. The text-to-image loss is:

$$\mathcal{L}_{t2i} = -\frac{1}{40} \sum_{j=1}^{40} \log \frac{\exp(S_{i^*(j),j})}{\sum_{i=1}^8 \exp(S_{i,j})}. \quad (32)$$

The total alignment objective is:

$$\mathcal{L} = \frac{1}{2} (\mathcal{L}_{i2t} + \mathcal{L}_{t2i}) + \mathcal{L}_{\text{KL}}. \quad (33)$$

The end-to-end training procedure for the proposed Native Mamba-2 + HEDO framework is formalized in Algorithm 1.

Algorithm 1 Training Procedure for Native Mamba-2 + HEDO

Require: Image-caption dataset $\mathcal{D}$, frozen backbones ViT-B/16, RoBERTa-base, epochs $E = 10$, batch size $B_{\text{img}} = 8, B_{\text{txt}} = 40$

Ensure: Trained projection layers $\mathbf{W}_v, \mathbf{W}_t$, HEDO parameters θ_{hedo} , Native Mamba-2 parameters θ_{ssd} , log-temperature τ

- 1: Initialize projection layers $\mathbf{W}_v, \mathbf{W}_t \in \mathbb{R}^{768 \times 128}$, HEDO parameters $\mathbf{W}_q, \mathbf{W}_p, \mathbf{A}, \gamma, \mathbf{W}_o$, Mamba-2 SSD weights, and $\tau \leftarrow \log(1/0.07)$
 - 2: **for** epoch = 1 **to** E **do**
 - 3: **for** each grouped batch $(\mathcal{B}_{\text{img}}, \mathcal{B}_{\text{txt}}) \sim \mathcal{D}$ **do**
 - 4: Extract frozen visual patch features $\mathbf{X}_v \leftarrow \text{ViT}(\mathcal{B}_{\text{img}}) \in \mathbb{R}^{8 \times 196 \times 768}$
 - 5: Extract frozen text token features $\mathbf{X}_t \leftarrow \text{RoBERTa}(\mathcal{B}_{\text{txt}}) \in \mathbb{R}^{40 \times L \times 768}$
 - 6: Project to latent dimension: $\mathbf{H}_v^{(0)} \leftarrow \mathbf{X}_v \mathbf{W}_v, \mathbf{H}_t^{(0)} \leftarrow \mathbf{X}_t \mathbf{W}_t \in \mathbb{R}^{8 \times \times \times 128}$
 - 7: Apply HEDO adapter: $\mathbf{H}_v \leftarrow \text{HEDO}(\mathbf{H}_v^{(0)}), \mathbf{H}_t \leftarrow \text{HEDO}(\mathbf{H}_t^{(0)})$ via (19)
 - 8: Process through Native Mamba-2 SSD blocks: $\mathbf{Y}_v \leftarrow \text{SSD}_v(\mathbf{H}_v), \mathbf{Y}_t \leftarrow \text{SSD}_t(\mathbf{H}_t)$
 - 9: Compute masked global mean pooling and ℓ_2 -normalization: $\mathbf{z}_v \leftarrow \text{Pool}(\mathbf{Y}_v), \mathbf{z}_t \leftarrow \text{Pool}(\mathbf{Y}_t) \in \mathbb{S}^{127}$
 - 10: Compute cosine similarity matrix: $\mathbf{S} \leftarrow \exp(\tau) \cdot (\mathbf{z}_v \mathbf{z}_t^T) \in \mathbb{R}^{8 \times 40}$
 - 11: Compute symmetric multi-positive InfoNCE loss: $\mathcal{L}_{\text{NCE}} \leftarrow \frac{1}{2} (\mathcal{L}_{i2t} + \mathcal{L}_{t2i})$ via (31)–(32)
 - 12: **if** AVSC enabled **then**
 - 13: Compute confidence-gated symmetric KL divergence $\mathcal{L}_{\text{KL}}$ via (28)
 - 14: Total objective: $\mathcal{L} \leftarrow \mathcal{L}_{\text{NCE}} + \mathcal{L}_{\text{KL}}$
 - 15: **else**
 - 16: Total objective: $\mathcal{L} \leftarrow \mathcal{L}_{\text{NCE}}$
 - 17: **end if**
 - 18: Backpropagate gradients $\nabla_{\theta} \mathcal{L}$
 - 19: Clip gradients: $\mathbf{g} \leftarrow \text{clip}(\mathbf{g}, \|\mathbf{g}\|_2 \leq 1.0)$
 - 20: Update trainable parameters θ using AdamW ($\eta = 10^{-4}$, weight decay 10^{-4})
 - 21: **end for**
 - 22: **end for**
 - 23: **return** Trained model parameters $\{\mathbf{W}_v, \mathbf{W}_t, \theta_{\text{hedo}}, \theta_{\text{ssd}}, \tau\}$
-

V. EXPERIMENTAL METHODOLOGY

A. Dataset

We evaluate on the canonical Flickr8k dataset [16], comprising 8,000 images, each paired with 5 human-written captions (6,000 training, 1,000 validation, and 1,000 test images).

B. Data Preprocessing

Images are resized to 224×224 and normalized using ImageNet statistics. Text is tokenized using the standard RoBERTa tokenizer with maximum sequence length truncated to 64 tokens.

C. Training Configuration

All models are trained for 10 epochs using AdamW with an initial learning rate of $\eta = 10^{-4}$ and weight decay 10^{-4} . Batch size is fixed to 8 unique images and 40 captions per iteration. Gradient clipping is enforced at a maximum ℓ_2 norm of 1.0.

D. Experimental Configurations

To evaluate selected individual and joint effects of HEDO, Mamba-2, MSA, and AVSC, we construct a controlled 8-configuration ablation benchmark, summarized in Table I. Because a full factorial exploration over four binary mechanisms would require $2^4 = 16$ configurations, our benchmark evaluates a targeted, structured subset of this space.

TABLE I
ABLATION CONFIGURATION MATRIX DEFINING ARCHITECTURAL VARIANTS.

Configuration ID	HEDO	Mamba-2	MSA	AVSC
baseline	×	×	×	×
mamba2_hedo_v2	✓	✓	×	×
mamba2_msa_v2	×	✓	✓	×
hedo_msa_v2	✓	×	✓	×
avsc_msa_v2	×	×	✓	✓
mamba2_avsc_v2	×	✓	×	✓
full_v2 (All)	✓	✓	✓	✓
hedo_avsc_v2	✓	×	×	✓

Table I defines the controlled 8-configuration ablation benchmark. While these eight configurations evaluate critical individual and joint additions (e.g., Mamba-2 + HEDO vs. baseline), they represent a structured subset of the 2^4 architectural space rather than an exhaustive full factorial matrix; consequently, they evaluate selected combinations rather than independently isolating every factor in complete isolation (such as Mamba-2 without HEDO, or MSA without Mamba-2/HEDO). Full V2 activates all four mechanisms simultaneously and serves as a compositional stress test rather than an assumption that all components are mutually compatible.

E. Evaluation Metrics

Retrieval performance is evaluated on the standard 1,000-image test set (1,000 images $\times$ 5,000 captions):

- **Image-to-Text (i2t) Retrieval:** Recall@1, Recall@5, Recall@10, and Median Rank (MedR).
- **Text-to-Image (t2i) Retrieval:** Recall@1, Recall@5, Recall@10, and Median Rank (MedR).
- **Mean Recall (MR):** The primary arithmetic mean across all six recall metrics:

$$\text{MR} = \frac{1}{6} \left(\text{R@1}_{i2t} + \text{R@5}_{i2t} + \text{R@10}_{i2t} + \text{R@1}_{t2i} + \text{R@5}_{t2i} + \text{R@10}_{t2i} \right). \quad (34)$$

F. Statistical Evaluation Protocol

For every configuration $c \in \{1, \dots, 7\}$, we compute the paired delta relative to the baseline across identical seeds $s \in \{42, 43, 44\}$:

$$\Delta_{c,s} = \text{MR}_{c,s} - \text{MR}_{\text{baseline},s}. \quad (35)$$

We report the sample mean $\bar{\Delta}_c$, sample standard deviation s_c , and two-tailed 95% Student- t confidence intervals:

$$\text{CI}_{95\%} = \left[\bar{\Delta}_c - t_{0.025,2} \frac{s_c}{\sqrt{3}}, \bar{\Delta}_c + t_{0.025,2} \frac{s_c}{\sqrt{3}} \right], \quad (36)$$

where $t_{0.025,2} = 4.303$. An architectural comparison relative to the baseline is classified as:

- **SUPPORTED (IMPROVEMENT):** $\bar{\Delta}_c > 0$ and $\text{CI}_{95\%}$ strictly excludes zero ($\text{CI}_{\text{low}} > 0$).
- **NOT SUPPORTED (NO STAT. DIFF.):** The null hypothesis of zero difference cannot be rejected as $\text{CI}_{95\%}$ contains zero ($\text{CI}_{\text{low}} \leq 0 \leq \text{CI}_{\text{high}}$).
- **CONTRADICTED (DEGRADATION):** $\bar{\Delta}_c < 0$ and $\text{CI}_{95\%}$ strictly excludes zero ($\text{CI}_{\text{high}} < 0$).

Because the analysis was designed as a predefined architectural audit with seven baseline comparisons, we report unadjusted two-sided 95% Student- t confidence intervals; multiplicity-adjusted inference is left for future expanded benchmarks.

G. Reproducibility and Audit Controls

To guarantee numerical reproducibility and prevent silent execution failures, the training harness incorporates strict hardware and numerical assertion gates:

- **Finite Tensor Monitor:** Every forward and backward pass asserts `isfinite(T)` across all intermediate activations, logit matrices, and gradient tensors.
- **Native Mamba-2 Verification:** Direct inspection confirms loading of official `mamba_ssm.modules.mamba2.Mamba2` GPU CUDA kernels.
- **Three-Seed Replication:** Every configuration is executed independently across seeds 42, 43, 44, generating 24 fully audited training runs. The methodology hash is recorded as `1753cfb7...1af54`¹.

The complete multi-seed experimental audit protocol is formalized in Algorithm 2.

VI. RESULTS

A. Overall Retrieval Performance

The empirical retrieval metrics across all 8 configurations are compiled in Table II and illustrated across recall thresholds in Fig. 2.

As reported in Table II, the baseline model achieves a Mean Recall of $37.16 \pm 0.57\%$, with Image-to-Text (i2t) R@1 of 16.53% and Text-to-Image (t2i) R@1 of 13.92%. Across the eight configurations, performance bifurcates cleanly into two distinct regimes: a stable regime where no statistically

¹Full 64-character SHA-256 methodology hash: `1753cfb7181e6355cd79bd6dd9a3ceb626b4de36f8dbcf0807591b89cd1af54`.

Algorithm 2 Multi-Seed Experimental Audit Protocol

Require: Configuration set $\mathcal{C} = \{c_0, c_1, \dots, c_7\}$, seeds $\mathcal{S} = \{42, 43, 44\}$, test dataset $\mathcal{D}_{\text{test}}$

Ensure: Paired deltas $\bar{\Delta}_c$, 95% confidence intervals $\text{CI}_{95\%}$, pre-specified verdicts

- 1: **for** each configuration $c \in \mathcal{C}$ **do**
- 2: **for** each random seed $s \in \mathcal{S}$ **do**
- 3: Set random seed s across PyTorch, NumPy, Python runtime
- 4: Assert GPU execution environment and official Mamba-2 CUDA kernel loading
- 5: Train model configuration c using Algorithm 1 with finite tensor monitoring
- 6: Evaluate test retrieval on $\mathcal{D}_{\text{test}}$: compute R@1, R@5, R@10, MedR for i2t and t2i
- 7: Compute Mean Recall: $\text{MR}_{c,s} \leftarrow \frac{1}{6} \sum \text{Recall}$ via (34)
- 8: Measure head/end-to-end latency, throughput, and peak VRAM
- 9: **end for**
- 10: Compute aggregate metrics: $\mu_c \leftarrow \text{mean}_s(\text{MR}_{c,s})$, $\sigma_c \leftarrow \text{std}_s(\text{MR}_{c,s})$
- 11: **if** $c \neq \text{baseline}$ **then**
- 12: Compute per-seed paired deltas: $\Delta_{c,s} \leftarrow \text{MR}_{c,s} - \text{MR}_{\text{baseline},s}$ via (35)
- 13: Compute sample mean $\bar{\Delta}_c$ and sample standard deviation s_c
- 14: Compute 95% Student- t confidence interval: $\text{CI}_{95\%} \leftarrow [\bar{\Delta}_c \pm t_{0.025,2} \frac{s_c}{\sqrt{3}}]$ via (36)
- 15: **if** $\text{CI}_{95\%} \subset (0, \infty)$ **then**
- 16: Set Verdict $\leftarrow$ **SUPPORTED**
- 17: **else if** $0 \in \text{CI}_{95\%}$ **then**
- 18: Set Verdict $\leftarrow$ **NO STAT. DIFF.**
- 19: **else**
- 20: Set Verdict $\leftarrow$ **CONTRADICTED**
- 21: **end if**
- 22: **end if**
- 23: **end for**
- 24: Generate tabular summaries (Tables II, III, IV) and statistical figures (Figs. 2–5)
- 25: **return** Audited benchmark results and statistical decisions

significant difference from the baseline was detected under the three-seed evaluation (Baseline and Mamba-2 + HEDO) and severe degradation regimes (MSA and AVSC). As visually evident in Fig. 2, this bifurcation is uniform across all recall thresholds.

B. HEDO + Native Mamba-2 Evaluation

Equipping the architecture with the Hamiltonian-inspired dissipative adapter and Native Mamba-2 sequence blocks (`mamba2_hedo_v2`) yields a Mean Recall of $37.31 \pm 0.68\%$, with i2t R@1 reaching $17.40 \pm 0.62\%$ and t2i R@1 reaching $14.20 \pm 0.58\%$. As shown in Table III, the mean paired delta is $+0.1533$ pp, with a 95% confidence interval of $[-1.4246, +1.7313]$ pp ($n = 3$ paired runs). Because the

TABLE II
COMPLETE ABLATION RETRIEVAL PERFORMANCE ON FLICKR8K ACROSS 8 CONFIGURATIONS (MEAN $\pm$ STANDARD DEVIATION ACROSS 3 SEEDS, $N = 24$).

Configuration	i2t R@1 (%)	i2t R@5 (%)	i2t R@10 (%)	t2i R@1 (%)	t2i R@5 (%)	t2i R@10 (%)	Mean Recall (%)	i2t MedR	t2i MedR	Verdict
Baseline	16.53 $\pm$ 0.70	43.27 $\pm$ 0.72	57.47 $\pm$ 1.14	13.92 $\pm$ 0.45	38.31 $\pm$ 0.38	53.45 $\pm$ 0.49	37.16 $\pm$ 0.57	7.67 $\pm$ 0.58	9.00 $\pm$ 0.00	<i>Reference</i>
Mamba-2 + HEDO	17.40 $\pm$ 0.62	43.47 $\pm$ 1.85	57.20 $\pm$ 1.10	14.20 $\pm$ 0.58	38.25 $\pm$ 0.47	53.35 $\pm$ 0.50	37.31 $\pm$ 0.68	7.67 $\pm$ 0.58	9.00 $\pm$ 0.00	No Stat. Diff.
Mamba-2 + MSA	9.77 $\pm$ 0.51	29.53 $\pm$ 0.95	42.90 $\pm$ 1.04	8.92 $\pm$ 0.47	28.22 $\pm$ 0.59	41.98 $\pm$ 0.75	26.89 $\pm$ 0.68	13.83 $\pm$ 0.76	14.67 $\pm$ 0.58	Contradicted
HEDO + MSA	9.47 $\pm$ 0.47	29.50 $\pm$ 1.47	42.87 $\pm$ 1.59	8.33 $\pm$ 0.42	27.45 $\pm$ 0.09	40.84 $\pm$ 0.84	26.41 $\pm$ 0.49	14.67 $\pm$ 0.58	15.33 $\pm$ 0.58	Contradicted
AVSC + MSA	1.47 $\pm$ 0.32	6.40 $\pm$ 1.87	11.50 $\pm$ 1.84	1.50 $\pm$ 0.26	6.51 $\pm$ 0.28	12.48 $\pm$ 0.76	6.64 $\pm$ 0.82	81.67 $\pm$ 5.84	61.33 $\pm$ 2.89	Contradicted
Mamba-2 + AVSC	1.80 $\pm$ 0.78	6.27 $\pm$ 1.10	11.87 $\pm$ 0.96	1.05 $\pm$ 0.32	5.55 $\pm$ 1.38	10.19 $\pm$ 2.74	6.12 $\pm$ 1.14	99.83 $\pm$ 21.50	72.33 $\pm$ 14.74	Contradicted
Full V2 (All)	1.47 $\pm$ 0.31	5.97 $\pm$ 0.47	10.73 $\pm$ 0.68	1.25 $\pm$ 0.29	5.60 $\pm$ 0.69	10.69 $\pm$ 1.09	5.95 $\pm$ 0.42	94.33 $\pm$ 6.66	69.67 $\pm$ 3.79	Contradicted
HEDO + AVSC	1.03 $\pm$ 0.12	4.93 $\pm$ 0.90	9.27 $\pm$ 1.05	1.01 $\pm$ 0.23	4.83 $\pm$ 1.15	9.20 $\pm$ 1.89	5.04 $\pm$ 0.84	103.17 $\pm$ 10.13	79.00 $\pm$ 13.23	Contradicted

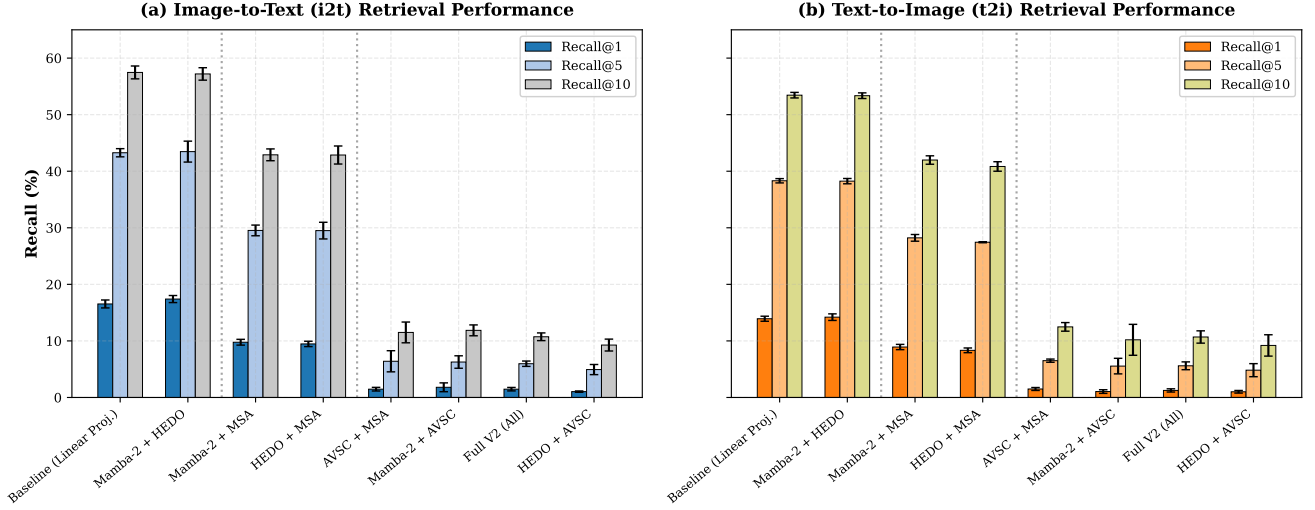


Fig. 2. Controlled ablation benchmark on Flickr8k across all 8 configurations (3 independent random seeds, $N = 24$ runs). (a) Image-to-Text (i2t) Recall@1, Recall@5, and Recall@10. (b) Text-to-Image (t2i) Recall@1, Recall@5, and Recall@10. Error bars denote ± 1 standard deviation. Note the striking bifurcation between the stable regime with no detected difference from baseline (Baseline and Mamba-2 + HEDO) and the severe degradation regimes (MSA and AVSC).

confidence interval encompasses zero, the hypothesis of a statistically significant improvement (H_1) is classified as **NOT SUPPORTED** (no statistically significant difference was detected under the present three-seed evaluation).

These results indicate that Native Mamba-2 combined with the HEDO dissipative adapter preserves baseline-level retrieval performance under the evaluated conditions, without observed numerical instability, while maintaining low computational overhead.

C. MSA Ablation Analysis

Configurations incorporating Multi-Scale State Aggregation (mamba2_msa_v2 and hedo_msa_v2) exhibit a sharp and consistent drop in retrieval capability. mamba2_msa_v2 attains a Mean Recall of $26.89 \pm 0.68\%$, representing a severe paired delta of -10.2711 pp (95% CI: $[-13.1018, -7.4404]$ pp). Similarly, hedo_msa_v2 achieves $26.41 \pm 0.49\%$ ($\Delta = -10.7489$ pp, 95% CI: $[-12.3816, -9.1161]$ pp). Both results strictly exclude zero and contradict Hypothesis 2 (H_2). Instead of enriching sequence representations, multi-scale chunk boundary and terminal token states substantially degrade retrieval quality.

D. AVSC Ablation Analysis

The most striking empirical outcome is that Adaptive Variational State Coupling is associated with severe align-

ment degradation. Across all four configurations containing AVSC (avsc_msa_v2, mamba2_avsc_v2, full_v2, and hedo_avsc_v2), Mean Recall drops precipitously to $5.04\% - 6.64\%$, consistent with dimensional collapse under the evaluated formulation.

The paired deltas are uniformly severe:

- **AVSC + MSA:** $\Delta = -30.52$ pp (95% CI: $[-32.49, -28.54]$ pp)
- **Mamba-2 + AVSC:** $\Delta = -31.04$ pp (95% CI: $[-35.26, -26.81]$ pp)
- **Full V2 (All):** $\Delta = -31.21$ pp (95% CI: $[-32.60, -29.81]$ pp)
- **HEDO + AVSC:** $\Delta = -32.11$ pp (95% CI: $[-35.61, -28.62]$ pp)

Full V2 (All), which simultaneously combines HEDO, Mamba-2, MSA, and AVSC, attains $5.95 \pm 0.42\%$ Mean Recall, indicating that the variational degradation dominates and nullifies any beneficial inductive biases. Correspondingly, Median Rank (MedR), which is 7.67 (i2t) and 9.00 (t2i) for Baseline and Mamba-2 + HEDO, blows up to $81.67 - 103.17$ (i2t) and $61.33 - 79.00$ (t2i) in AVSC configurations. These outcomes contradict Hypothesis 3 (H_3).

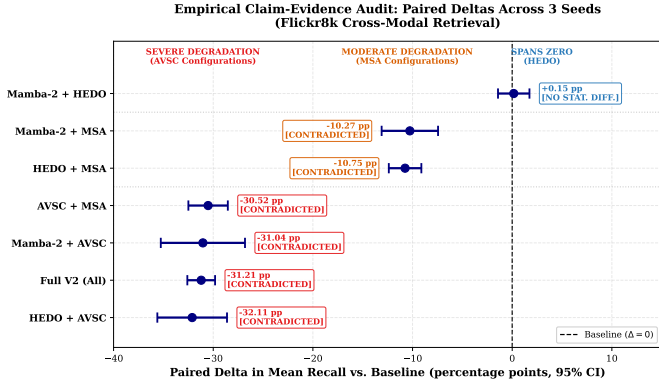


Fig. 3. Forest plot of paired deltas in Mean Recall relative to the baseline across 3 seeds. Points denote sample means; error bars denote 95% Student- t confidence intervals. The 95% confidence interval for Mamba-2 + HEDO spans zero (indicating no statistically significant difference from the baseline under this sample size), while all other configurations exhibit severe degradation.

E. Statistical Comparison

The detailed paired deltas across individual seeds and confidence intervals are reported in Table III and visualized as a forest plot in Fig. 3.

The forest plot in Fig. 3 visually summarizes the paired delta distributions across the $n = 3$ paired runs. The 95% confidence interval for Mamba-2 + HEDO ($[-1.42, +1.73]$ pp) spans zero; therefore, no statistically significant difference from the baseline was detected under the three-seed evaluation. In contrast, all MSA and AVSC configurations lie completely to the left of zero, indicating that the observed degradations are consistent across the evaluated seeds.

F. Computational Efficiency

Table IV and Fig. 4 quantify the computational and memory characteristics of all configurations.

Mamba-2 + HEDO requires 597,787 trainable parameters, adding 132,610 trainable parameters (+28.5% relative to the baseline trainable parameter budget of 465,177), while increasing the total parameter count by only 0.063% because the 210.6M backbone parameters remain frozen. Relative to the baseline linear projection, peak training VRAM increases from 1,042.31 MB to 1,086.15 MB (+43.84 MB, a 4.2% increase), head latency increases from 4.19 ± 0.13 ms to 5.67 ± 0.13 ms (+35.3%), and end-to-end latency increases from 7.64 ± 0.23 ms to 8.39 ± 0.08 ms (+9.8%). Despite these moderate overheads, head throughput remains high at $7,054.12 \pm 161.35$ pairs/s ($4,765.93 \pm 45.65$ pairs/s end-to-end), supporting Hypothesis 4 (H_4) by confirming practical computational throughput. As shown in Fig. 4, Mamba-2 + HEDO lies on the observed accuracy-throughput Pareto frontier among the eight evaluated configurations.

G. Multi-Seed Stability

Fig. 5 illustrates the stability across seeds 42, 43, and 44. The standard deviation in Mean Recall across seeds is small ($\leq 0.68\%$ for the top configurations), indicating consistency across the evaluated seeds, although larger seed ensembles are

required to quantify sensitivity to random initialization more precisely.

VII. MECHANISTIC ANALYSIS AND DISCUSSION

The clear bifurcation between the stability of HEDO and the failure of MSA and AVSC demands thorough mechanistic analysis.

A. Interpretation of HEDO Stability

Unlike AVSC and MSA, HEDO maintains performance with no statistically significant difference detected from the baseline under the three-seed evaluation. This stability is governed by three structural properties formalized in Section IV-D:

- 1) **Exact Skew-Symmetry:** By parameterizing $\mathbf{J} = (\mathbf{A} - \mathbf{A}^T)/\sqrt{d}$ via (12), HEDO enforces $\mathbf{v}^T \mathbf{J}^T \mathbf{v} = 0$ identically for all $\mathbf{v}$. The skew-symmetric component contributes no direct change to the quadratic Hamiltonian energy in the continuous-time formulation.
- 2) **Positive Diagonal Damping:** The damping matrix $\mathbf{R} = \text{diag}(0.05 + 0.95\sigma(\gamma))$ via (13) guarantees that energy is dissipated monotonically ($\mathcal{H} \leq 0$ by Proposition 1), preventing activation blow-up or exploding gradients.
- 3) **Residual Damping Scaling:** The residual update $\mathbf{x}_{\text{out}} = \mathbf{x} + 0.10 \cdot \mathbf{W}_o \text{LayerNorm}(\mathbf{q}_{\text{new}} + \mathbf{p}_{\text{new}})$ in (19) ensures that the adapter acts as a near-identity perturbation at initialization, maintaining the pre-trained manifold geometry of the frozen backbones.

B. Why MSA Degrades Retrieval: Recurrent Pooling Asymmetry

Multi-Scale State Aggregation attempted to enrich embeddings by concatenating chunk-boundary states $\bar{\mathbf{y}}_{\text{boundary}}$ and the terminal valid state $\mathbf{y}_{\text{last}}$ with the global mean state $\bar{\mathbf{y}}_{\text{global}}$. This resulted in an immediate ≈ 10.5 pp drop in Mean Recall.

We hypothesize that the degradation arises from the interaction between recurrent positional asymmetry and the selected pooling scheme:

- 1) **Recurrent Positional Asymmetry:** In Mamba-2, state transitions are computed sequentially:

$$\mathbf{h}_t = \mathbf{A}_t \mathbf{h}_{t-1} + \mathbf{B}_t \mathbf{x}_t. \quad (37)$$

The terminal valid state $\mathbf{y}_{\text{last}} = \mathbf{Y}_{B, L_{\text{valid}}-1}$ reflects the cumulative recency bias of the sequential recurrence. In language, it corresponds to the final valid token, whose semantic or positional role may differ substantially across captions. In vision, the final token corresponds to the bottom-right image patch (14×14 grid). Concatenating $\mathbf{y}_{\text{last}}$ introduces severe positional asymmetry into the multimodal embedding.

- 2) **Chunk Boundary Arbitrariness:** Chunk-boundary states $\mathbf{Y}_{\text{boundary}}$ capture the internal recurrent state at arbitrary 16-token intervals. Because visual entities and linguistic phrases do not align with uniform 16-token boundaries, these states represent arbitrary recurrent snapshots rather than coherent semantic constituents.

TABLE III
PAIRED DELTA ANALYSIS RELATIVE TO BASELINE ACROSS 3 INDEPENDENT SEEDS ($N = 3$ PAIRED RUNS PER CONFIGURATION).

Comparison	Per-Seed Deltas (%)	Mean Δ (pp)	SD Δ (pp)	95% CI (pp)	CI Excludes 0?	Predefined Audit Verdict
Mamba-2 + HEDO vs. Baseline	{42 : -0.27, 43 : -0.15, 44 : +0.88}	+0.1533	0.6352	[-1.4246, +1.7313]	False	NO STAT. DIFF. (Not Supported)
Mamba-2 + MSA vs. Baseline	{42 : -11.26, 43 : -9.02, 44 : -10.53}	-10.2711	1.1395	[-13.1018, -7.4404]	True	CONTRADICTED (-10.27 pp)
HEDO + MSA vs. Baseline	{42 : -11.12, 43 : -9.99, 44 : -11.14}	-10.7489	0.6573	[-12.3816, -9.1161]	True	CONTRADICTED (-10.75 pp)
AVSC + MSA vs. Baseline	{42 : -31.01, 43 : -30.94, 44 : -29.60}	-30.5156	0.7967	[-32.4947, -28.5364]	True	CONTRADICTED (-30.52 pp)
Mamba-2 + AVSC vs. Baseline	{42 : -32.98, 43 : -29.83, 44 : -30.30}	-31.0378	1.7008	[-35.2627, -26.8129]	True	CONTRADICTED (-31.04 pp)
Full V2 (All) vs. Baseline	{42 : -31.55, 43 : -30.56, 44 : -31.51}	-31.2067	0.5604	[-32.5987, -29.8146]	True	CONTRADICTED (-31.21 pp)
HEDO + AVSC vs. Baseline	{42 : -33.71, 43 : -31.05, 44 : -31.58}	-32.1133	1.4073	[-35.6092, -28.6174]	True	CONTRADICTED (-32.11 pp)

TABLE IV
COMPUTATIONAL EFFICIENCY, TRAINABLE PARAMETER FOOTPRINT, LATENCY, VRAM, AND THROUGHPUT METRICS (MEAN ACROSS 3 SEEDS).

Configuration	Trainable Params	Total Params	Peak VRAM (MB)	Train Time (s)	Head Latency (ms)	E2E Latency (ms)	Head Throughput (p/s)	E2E Throughput (p/s)
Baseline	465,177	211,087,873	1,042.31	278.44 $\pm$ 4.06	4.19 $\pm$ 0.13	7.64 $\pm$ 0.23	9,549.93 $\pm$ 291.85	5,240.09 $\pm$ 160.95
Mamba-2 + HEDO	597,787	211,220,483	1,086.15	362.76 $\pm$ 4.02	5.67 $\pm$ 0.13	8.39 $\pm$ 0.08	7,054.12 $\pm$ 161.35	4,765.93 $\pm$ 45.65
Mamba-2 + MSA	565,273	211,187,969	1,045.36	306.34 $\pm$ 2.27	4.49 $\pm$ 0.14	7.86 $\pm$ 0.19	8,916.05 $\pm$ 280.69	5,089.16 $\pm$ 118.87
HEDO + MSA	697,883	211,320,579	1,088.07	399.01 $\pm$ 4.32	5.80 $\pm$ 0.13	9.06 $\pm$ 0.32	6,902.35 $\pm$ 149.39	4,418.65 $\pm$ 152.66
AVSC + MSA	631,321	211,254,017	1,046.63	367.30 $\pm$ 2.19	5.70 $\pm$ 0.28	9.19 $\pm$ 0.36	7,029.26 $\pm$ 332.78	4,354.75 $\pm$ 166.50
Mamba-2 + AVSC	531,225	211,153,921	1,043.18	329.82 $\pm$ 2.51	5.16 $\pm$ 0.35	8.48 $\pm$ 0.30	7,773.80 $\pm$ 514.86	4,718.49 $\pm$ 163.03
Full V2 (All)	763,931	211,386,627	1,089.33	463.06 $\pm$ 1.49	7.07 $\pm$ 0.45	10.68 $\pm$ 0.39	5,671.25 $\pm$ 375.35	3,749.01 $\pm$ 139.28
HEDO + AVSC	663,835	211,286,531	1,087.42	427.66 $\pm$ 3.20	6.16 $\pm$ 0.25	9.74 $\pm$ 0.51	6,500.59 $\pm$ 273.53	4,112.24 $\pm$ 209.50

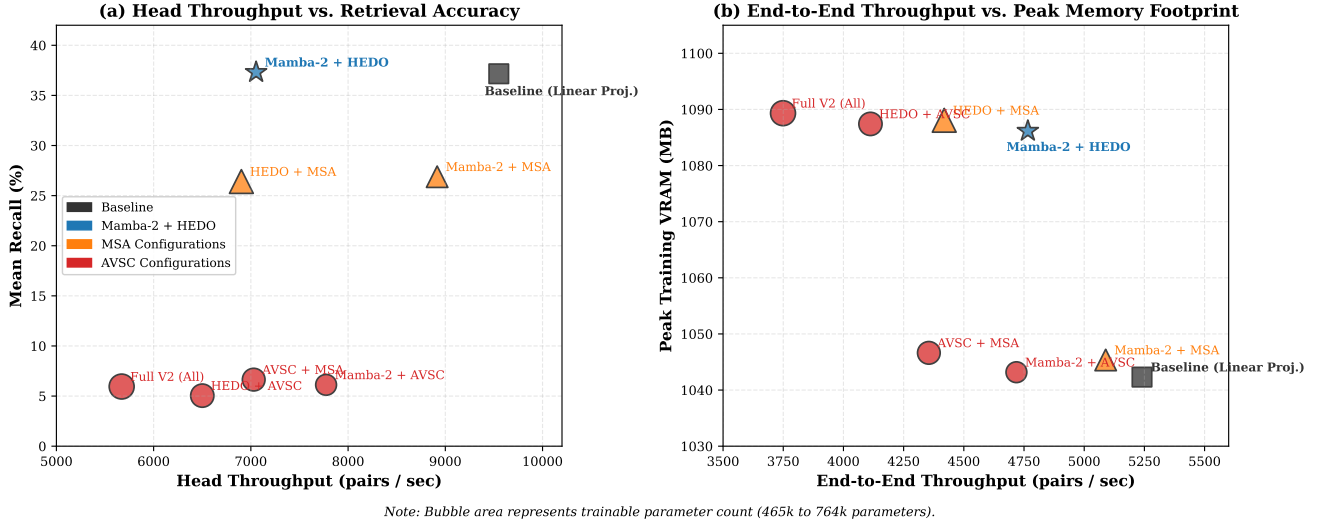


Fig. 4. Efficiency vs. accuracy tradeoffs across all 8 configurations. (a) Head throughput (pairs/s) vs. Mean Recall (%). (b) End-to-end throughput (pairs/s) vs. Peak Training VRAM (MB). Bubble area is proportional to the trainable parameter count (465k to 764k). Mamba-2 + HEDO lies on the observed accuracy-throughput Pareto frontier among the eight evaluated configurations, retaining baseline-level retrieval accuracy at high throughput and low memory.

3) **Order-Agnostic Mean Pooling:** Conversely, uniform masked mean pooling $\bar{\mathbf{y}}_{\text{global}} = \frac{\sum_{i=1}^L m_i \mathbf{y}_i}{\sum_{i=1}^L m_i}$ computes an unweighted barycenter over all contextualized token representations, providing an order-agnostic aggregation of the contextualized sequence states suitable for contrastive alignment.

C. Mechanistic Interpretation of AVSC Degradation

The severe retrieval degradation observed across all AVSC configurations ($\Delta \approx -30.5$ to -32.1 pp, Mean Recall $\approx 5.0\% - 6.6\%$, MedR ≥ 60) demands careful mechanistic examination. Rather than a formal mathematical theorem of collapse, we provide a conservative mechanistic interpretation based on the competing geometric constraints of the objective functions.

1) **Competing Geometric Pressures:** In the evaluated architecture, two distinct objectives operate at different stages of representation processing:

1) **Hyperspherical Contrastive Repulsion:** The multi-positive InfoNCE loss operates on final ℓ_2 -normalized embeddings $\mathbf{z}_v, \mathbf{z}_t \in \mathbb{S}^{d-1}$. As established in contrastive representation theory [14], [17], InfoNCE balances positive attraction with negative repulsion. The repulsive gradient across negative pairs,

$$\nabla_{\mathbf{z}_i} \mathcal{L}_{\text{rep}} = \sum_{j \in \mathcal{N}(i)} \frac{\exp(\tau \mathbf{z}_i^T \mathbf{z}_j)}{\sum_k \exp(\tau \mathbf{z}_i^T \mathbf{z}_k)} \tau \mathbf{z}_j, \quad (38)$$

pushes non-matching embeddings apart, promoting uniform dispersion and high effective rank across the unit hypersphere $\mathbb{S}^{d-1}$.

2) **Intermediate Distribution Matching:** Concurrently, the AVSC objective penalizes the symmetric Kullback-

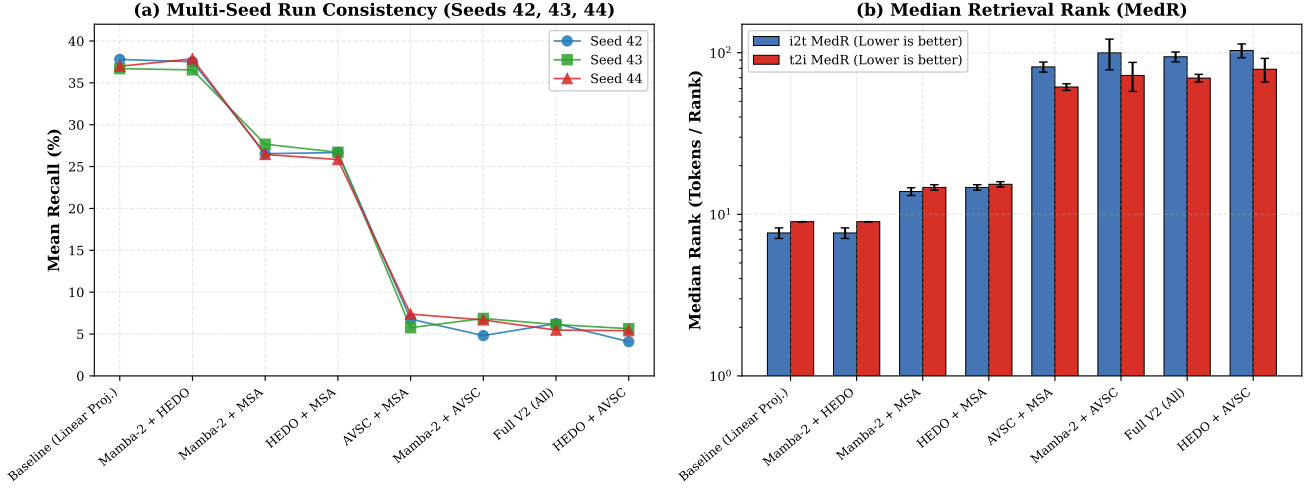


Fig. 5. Multi-seed run stability and retrieval rank distributions. (a) Mean Recall across seeds 42, 43, and 44, showing consistent performance across the three evaluated random seeds. (b) Log-scale Median Retrieval Rank (MedR) for image-to-text and text-to-image matching; lower values denote superior retrieval ranking precision.

Leibler divergence $\mathcal{D}_{\text{KL}}^{\text{sym}}(q_v, q_t)$ between diagonal Gaussian posteriors $q_v = \mathcal{N}(\mu_v, \Sigma_v)$ and $q_t = \mathcal{N}(\mu_t, \Sigma_t)$ parameterized by linear heads on intermediate representations $\mathbf{h}_v, \mathbf{h}_t \in \mathbb{R}^d$ via (25)–(26).

2) *Optimization Conflict and Representation Failure*: The symmetric KL divergence constrains the intermediate vision and text Gaussian posteriors toward distributional agreement in both their means and variances. However, because this constraint is applied to intermediate representations while the final retrieval objective operates on normalized embeddings, the two objectives impose competing geometric constraints on the representation space.

In particular, contrastive learning requires preserving fine-grained cross-modal discrepancies to discriminate among 40 in-batch candidates, whereas the intermediate Gaussian distribution-matching penalty penalizes discrepancy between paired representations regardless of the global contrastive configuration. Under the evaluated formulation and hyperparameters ($\lambda_{\text{KL}} = 10^{-3}$, clamp range $[-6.0, 2.0]$), the resulting optimization dynamics coincide with severe retrieval degradation.

While these empirical results—including near-zero Recall@1 (1.0% – 1.8%) and drastic median rank inflation (61 – 103)—are consistent with a collapse-like representation failure under the evaluated formulation, the present experiments establish this empirical association but do not by themselves prove that symmetric KL uniquely causes dimensional collapse. Conclusive verification of dimensional collapse would require empirical tracking of the singular value spectrum and effective embedding rank during training, which we identify as an important direction for future investigation.

D. Efficiency–Accuracy Tradeoff

Fig. 4 illustrates the Pareto frontier across all configurations. The baseline linear projection achieves the highest throughput (9,550 pairs/s) and lowest latency (4.19 ms). Mamba-2 +

HEDO retains baseline-level retrieval accuracy (37.31%) with moderate computational overhead, sustaining 7,054 pairs/s throughput with an incremental VRAM cost of only 43.8 MB (1,086 MB vs. 1,042 MB) and lying on the observed accuracy-throughput Pareto frontier. In contrast, all configurations incorporating AVSC or MSA incur substantial computational overhead without performance benefits.

E. Practical Design Implications

Based on our empirical and mathematical findings, we propose three concrete design principles for multimodal state-space architectures:

- 1) **Rely on Global Isotropic Pooling**: Avoid concatenating recurrent terminal states or intermediate chunk boundaries in bidirectional retrieval tasks. Masked mean pooling over all contextualized SSD sequence tokens provides superior semantic invariance.
- 2) **Exercise Extreme Caution with Intermediate Variational Alignment**: Do not superimpose cross-modal variational KL divergence objectives onto hyperspherical InfoNCE contrastive heads. The conflicting gradient dynamics can promote dimensional collapse under the evaluated formulation.
- 3) **Structure Continuous Adapters with Dissipative Constraints**: When physics-inspired continuous-state adapters are deployed, exact skew-symmetry ($\mathbf{J} = -\mathbf{J}^T$) and strictly positive damping ($\mathbf{R} > 0$) are essential to guarantee energy boundedness and numerical stability.

VIII. LIMITATIONS

While our findings provide rigorous empirical and mathematical insights, several limitations should be noted:

- **Dataset Scope**: The benchmark is conducted on the canonical Flickr8k dataset. While Flickr8k provides a controlled setting with 5 captions per image, evaluating

on larger-scale datasets (e.g., MS-COCO, Conceptual Captions) would assess scalability to larger vocabularies.

- **Backbone Scope:** Both vision (ViT-B/16) and language (RoBERTa-base) encoders were kept strictly frozen. While freezing isolates the sequence models, end-to-end fine-tuning may exhibit different optimization interactions.
- **Number of Seeds:** The benchmark evaluated three random seeds per configuration ($N = 24$ total runs). Although paired Student- t confidence intervals can be computed from these three paired observations, the small sample size limits statistical power; larger seed ensembles would provide more precise estimates.
- **Architecture Scope:** The study evaluated a specific Mamba-2 parameterization ($d_{\text{model}} = 128, d_{\text{state}} = 64, Q = 16$). Exploring varying chunk sizes and state dimensions remains an area for future work.
- **Mechanistic Analysis Scope:** Our mechanistic interpretation identifies competing geometric pressures between intermediate distribution matching and hyperspherical contrastive repulsion, which is consistent with the observed retrieval failure; however, direct covariance-spectrum diagnostics (such as singular-value decay and effective rank tracking) were not measured and are required to establish dimensional collapse conclusively.

IX. CONCLUSION

This paper presented an audited, multi-seed empirical investigation of continuous-state and variational inductive biases in Native Mamba-2 cross-modal alignment. Through a controlled 8-configuration ablation benchmark across 24 training runs on Flickr8k ($n = 3$ paired runs per configuration), our findings establish that Native Mamba-2 combined with the Hamiltonian dissipative adapter (HEDO) preserves baseline-level retrieval performance under the evaluated Flickr8k, frozen-backbone, three-seed setting ($37.31 \pm 0.68\%$ vs. $37.16 \pm 0.57\%$ Mean Recall, paired $\Delta = +0.15$ pp, 95% CI $[-1.42, +1.73]$ pp, no statistically significant difference detected), while maintaining high throughput (7,054 pairs/s) and low memory (1,086 MB peak VRAM). In contrast, Multi-Scale State Aggregation (MSA) and Adaptive Variational State Coupling (AVSC) configurations exhibit statistically significant degradation ($\Delta \approx -10.5$ pp and $\Delta \approx -31$ pp, respectively). For MSA, this degradation is consistent with recurrent positional asymmetry corrupting isotropic pooling. For AVSC, the empirical results are consistent with a representation-collapse mechanism arising from competing geometric constraints between intermediate distribution matching and hyperspherical contrastive repulsion, although direct covariance-spectrum diagnostics are required to establish dimensional collapse conclusively. These findings provide foundational empirical clarity and actionable architectural guidelines for integrating state space duality into multimodal foundation models.

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